

of semiconductor devices due to Covid, especially of processors, is leading to lower sales, even though products are in demand. One question people may ask is, if we don't have the foundries, how it will improve the supply chain? The answer is in the business model. Foundries like TSMC work based on committed contracts, and if you have signed up for a fixed number of devices and you have honoured the commitments previously, they do supply as per the agreement.

3. Third is having own micro-processor helps to train the designers in design and also ensures the knowledge is captured and used. This design knowledge is very critical for the defence and smart city applications, which are critical and need to be secured as they impact the society.

In addition, an Indian processor would also bring the cost down and reduce the development time as the processor proliferates. Above all, it would save the country from obsolescence of the processors in a short time, leading to support and new design effort issues. Having Indian processor allows the product to be obsoleted based on a planned strategy instead of panic mode reaction. (The panic is despite the IC vendors giving proper obsolescence notice as well as last time buy options, which product companies tend to ignore!)

How India can have own processor

With the semiconductor supply chain crisis hitting every domain, the issue of supply chain security has come to fore and shown how critical it is to secure a supply chain. Having learnt, it is high time that India has its own microprocessor with a secured supply chain. With most processors being supplied by global vendors, India must have own processor that can use

the RISC-V architecture as the best option.

In this regard, IIT Madras has been a pioneer and has developed two processors, RISECREEK and MOUSHIK. IITM's team has been successful in the conversion of the RISC-V architecture into ASICs as well as FPGA based processors. Since RISC-V defines the instruction architecture, speed and other aspects can be decided by the design teams offering flexibility.

One advantage is that the architecture is not technology specific, and designers can target different fabrication technologies to get different speeds. One of the biggest advantages of the IITM's design is the availability of operating systems for these processors. IITM team has ported both Linux and Free RTOS on to these processors. They have used open source tools for the software development, making the ecosystem truly open source.

Yet, one challenge that designers face will be that the tools being open source will not have the integration of the commercial IDE provided by the regular vendors. The tools are a little difficult to use. But this will change once the adoption of RISC-V picks up. Newer versions will address the security and other aspects needed for trusted devices.

Key to success in having an Indian processor lies in how quickly commercial device developers adopt the RISC-V architecture and develop different flavours of the processor. While it is easier said than done, need for the Indian processor will only increase as the time passes. Some of the challenges that need to be overcome are:

1. A processor development needs a huge investment to get a working processor taped out commercially (through outsourced foundry services). Waiting for a foundry to come up is not a viable option as foundries with advanced

technologies are expensive.

2. However, India should focus on setting up OSAT (outsourced semiconductor assembly & test) facilities which convert the processed wafers with ICs into individual devices. OSAT facilities are less expensive and more manual work oriented, so they can be a good option as a business model too. Also OSAT facilities are device-independent and very close to the business model of EMS (electronic manufacturing services) vendors.

3. Creation of software tools, libraries, domain-specific applications (like avionics, automotive, and medical, which need reliable software modules) and reference hardware and circuits for the designers to try their ideas before they commit to the product design.

4. Training designers, especially students, in the RISC-V architecture and its ecosystem so that when they start designing, they have the knowledge needed for developing products. Today, students still study 8085 and 8086 processors, which are obsolete and not in production. At the same time teaching them RISC-V without the devices being available is a wasted effort.

5. Startups who work on RISC-V based system to kick start the next wave of products.

From the support point of view, government should support the effort by incentivising the designs based on RISC-V and mandating the use of these processors in critical government departments like defense, citizen's benefit programmes like Aadhaar, and backing devices.

To conclude, with so much of hype around an Indian processor, we need to remember that for a processor to become successful it needs support at multiple levels. Harping on a foundry is not going to solve the problem and creating an ecosystem is a must. **EFY**